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ITEM 1

a Rate = $k[A]^n[B]^m$ L

Order of reaction with respect to B

$\left(\frac{8.00\times10^{-1}}{4.00\times10^{-1}}\right)^m = \frac{3.00\times10^{-3}}{1.50\times10^{-3}}$ Im

$m = 1$ Im

Therefore, the order of reaction with respect to B is 1.

Order of reaction with respect to A

$\left(\frac{5.00\times10^{-1}}{2.50\times10^{-1}}\right)^n = \left(\frac{6.00\times10^{-3}}{1.5\times10^{-3}}\right)$ Im

$A = 2^n$

$2^2 = 2^n$

$n = 2$ Im

Therefore the order of reaction with respect to A is 2.

From the rate law

Rate = $k[A]^2[B]^1$ L

Taking Experiement 1

$1.50\times10^{-3} = k(2.50\times10^{-1})^2(4.00\times10^{-1})^1$

$1.50\times10^{-3} = 0.025k$ Im

$K = \frac{1.50\times10^{-3}}{0.025}$

$K = 0.06\ mol^{-2}dm^6s^{-1}$ Im

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Using the thermodynamic data;

$\Delta H_{rxn} = \sum \Delta H_f^{\ominus}(\text{Products}) - \sum \Delta H_f^{\ominus}(\text{Reactants})$ ✓L

$= [\Delta H_f^{\ominus}(\text{products}) + \Delta H_f^{\ominus}(H_2)] - [2\Delta H_f^{\ominus}(A) + \Delta H_f^{\ominus}(B)]$ ✓Im

$= [-650.00 + 0.00] - [2(-950) + (-105.0)]$

$= -650.00 - [-1900 - 105.0]$ ✓Im

$= -3550 kJmol^{-1}$ ✓Im

Conclusion.

Since the calculated rate constant value, K for the reaction is $0.06 mol^{-2} dm^6 s^{-1}$, thus less than $0.08 mol^{-2} dm^6 s^{-1}$ and the enthalpy of reaction obtained is $-355.0 kJmol^{-1}$ thus ~~less~~ greater than $-300.0 kJmol^{-1}$ reaction is exothermic therefore can be used to generate electricity and production efficiency for the process is high.

b) Factors

Increasing the concentration of reactants, A and B.

Increase in the concentration of reactants increases the chances of collisions forming more hydrogen.

Temperature

Increase in temperature increases the kinetic energy of the reactants increase the chances of successful collision between reacting spices.

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Item two

a) From the graph:

P_z at 97°C = 0·62atm ✓ Im

P_{H_2O} at 97°C = 0·384 atm ✓ Im

M_z = 4·2g

M_{water} = 1·1g

Molar mass of water = 18g

$\frac{M_z}{M_w} = \frac{P_z \times M_z}{P_{water} \times M_{water}}$ ✓ Im

$\frac{4·2}{1·1} = \frac{0·620 \times M_z}{0·384 \times 18}$ ✓ Im

$M_z = \frac{4·2 \times 6·912}{1·1 \times 0·620}$

$= 42·6g$

$\approx 43g$ ✓ Im

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Item two.

b) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

	$N_{2(g)}$	$3H_{2(g)}$	$2NH_{3(g)}$
initial moles	2·5	7·5	0
Moles formed	x	$3x$	$2x$ ✓ Im
Moles at equilibrium	$2·5-X$	$7·5-3X$	$2x$ ✓ Im

Value of x

$2X = 1·6$

$X = 0·8$ moles ✓ Im

Moles of $H_{2(g)}$

$7·5-(3\times0·8)$

5·1 moles ✓ Im

Moles of $N_{2(g)}$

$2·5-X$

$2·5-0·8$

1·7 moles ✓ Im

Concentration

$[NH_3] = \frac{1·6}{4} = 0·4M$ ✓ Im.

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$[H_2] = 5.1/4 = 1.275M$ ✓ Im

$[N_2] = 1.7/4 = 0.425M$ ✓ Im

$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$ ✓ L

$= \frac{(0.400)^2}{0.425 \times (1.275)^3}$ ✓ Im

$= 0.182 mol^{-2} dm^6$ ✓ Im.

The molecular mass of Z obtained is 43.09, therefore above the required value of 32.09. And the Kc value obtained is $0.182 mol^{-2} dm^6$, greater then $0.15 mol^{-2} dm^6$, the required Kc value. ✓ Dc

Therefore since the molecular mass obtained is above 32g and the Kc value obtained is above $0.15 mol^{-2} dm^6$ hence the company has a high production efficiency. ✓ Dc.

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* Item; Data Analysed; Dc
Dc = Data

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item 2.

Pressure ✓F

Increase in pressure shifts equilibrium from left to right since forward reaction occurs by a decrease in volume. Nitrogen will react with hydrogen to form ammonia, thus increasing the yield of ammonia gas because there are more gas molecules in a given volume, molecules are closer together ~~Hence~~ hence there are more chances of successful collisions between particles. ✓e. Expln.

Temperature ✓F

Decrease in temperature favours the forward since its exothermic, equilibrium shifts from left to right ✓e

Any two factors explained (2Fe)

2Fe = max score

Real-life applications

- Steam distillation; Extraction of oils, perfumes, food flavorings agents from plants.
- Chemical equilibrium; Applied in the manufacture of Sulphuric acid during the Contact process.
- Ammonia production during the Haber process.
- Nitric acid manufacture.

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Item three

d) The melting point of the chlorides decreases from sodium chloride to magnesium chloride, aluminium chloride and silicon tetrachloride.

Sodium chloride and magnesium chloride have giant ionic structures held by strong ionic bonds which require high amount of energy to break. The decrease in melting point from sodium chloride to magnesium chloride is because the magnesium ion has a smaller ionic radius, higher charge density and higher polarising power than sodium ion; making magnesium chloride less ionic than sodium chloride.

Aluminium chloride has a lower melting point than magnesium chloride. This is because the aluminium ion in aluminium chloride has a smallest ionic radius, highest polarising power and highest charge density making aluminium chloride predominantly covalent; Silicon tetrachloride is covalent in nature, held by weak van der waals forces of attraction since it has a simple molecular structure. Therefore requires a low amount of energy to break thus low melting point.

Fe = 15 points. (15 scoring points).

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Reactions of the chlorides with:

Water:

Sodium chloride and magnesium chloride dissolves in water to form neutral solutions containing sodium ions, magnesium ions, chloride ions respectively.

$NaCl_{(s)} + (aq) \longrightarrow Na^{+}_{(aq)} + Cl^{-}_{(aq)}$ ✓ Eq.

$MgCl_{2(s)} + (aq) \longrightarrow Mg^{2+}_{(aq)} + 2Cl^{-}_{(aq)}$ ✓ Eq.

Eq; Equation

Aluminium chloride hydrolyses in water to form a white precipitate of aluminium hydroxide and hydrochloric acid

$AlCl_{3(s)} + 3H_2O_{(l)} \longrightarrow Al(OH)_{3(s)} + 3HCl_{(aq)}$ ✓ Eq

(OR) $Al^{3+}_{(aq)} + 3H_2O_{(l)} \longrightarrow Al(OH)_{3(s)} + H^{+}_{(aq)}$ ✓

Silicon tetrachloride hydrolyses in water to form silicon(IV) oxide and hydrochloric acid.

$SiCl_{4(l)} + 2H_2O_{(l)} \longrightarrow SiO_{2(s)} + 4HCl_{(aq)}$ ✓

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Task (c) item 3.

Sodium chloride is used in food preservation, chemical manufacture of chlorine gas and sodium hydroxide A

Sodium chloride when disposed improperly in the soil, because high concentration of sodium and chloride ions interfere with plant water uptake and cause toxicity leading to poor growth or death of soil living organisms. D

Mitigated by avoiding excessive discharge of sodium chloride into freshwater/soil. M

Magnesium chloride is used production of magnesium, cement and dust control. A Magnesium chloride when disposed improperly in the soil and water increases their salinity and harm both water and soil organisms. D

Mitigated by preventing unnecessary release of magnesium chloride into water bodies. M

Aluminium chloride is important catalyst in organic industries especially in Friedel-Crafts reactions. A

Aluminium chloride hydrolyses in water to form acidic solutions that low be soil and water pH thus death of some organisms, M

Mitigated by use of baric oxide to neutralise the acidic soils. D

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item 3 (c).

Silicon tetrachloride is important in electronics and optinical-fibre manufacture. ✓ A

Silicon tetrachloride highly hydrolyzes in water producing hydrogen chloride fumes, which are ✓ Δ corrosive to skin, eyes and respiratory system

Mitigated by use of sealed containers when handling, wear gloves, goggles and also ✓ M strictly keep silicon tetrachloride away from water

A = Application

Δ = Danger.

M = mitigation.

Ae = application Explained.

2Ae = max score

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item 4

(a)

Electronegatively decreases from Fluorine to iodine. This is because from fluorine to iodine, nuclear charge increases, screening effect also increases, because an extra energy level completely filled with electrons is added from element to element. Increase in screening effect outweighs increase in nuclear charge. Effective nuclear charge decreases, atomic radius increases and bonding electrons experience greater replusion than nuclear attraction.

Fe = 08 scoring points.

Reactions of Fluorine and Iodine with water.

Chlorine disproportionates in water to form chloric(i) acid and hydrochloric acid.

$Cl_{2(g)} + H_2O_{(l)} \rightleftharpoons HOCl_{(aq)} + HCl_{(aq)}$ Eq

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item 4 (b):

Fluorine oxidises with cold water to form oxygen and hydrofluoric acid.

$2F_{2(g)} + 2H_2O_{(l)} \longrightarrow 4HF_{(aq)} + O_{2(g)}$

Reaction with sodium hydrogen solution:
Fluorine reacts with cold dilute sodium hydroxide solution to form oxygen difluoride, sodium fluoride and water, the type of reaction is a redox reaction

$2F_{2(g)} + 2NaOH_{(aq)} \longrightarrow 2NaF_{(aq)} + OF_{2(g)} + H_2O_{(l)}$

Chlorine disproportionates in cold dilute sodium hydroxide solution to form sodium chloride, sodium chlorate(I) and water

$Cl_{2(g)} + 2NaOH_{(aq)} \longrightarrow NaCl_{(aq)} + NaClO_{(aq)} + H_2O_{(l)}$

Fluorine reacts with hot concentrated sodium hydroxide to form oxygen gas, sodium fluoride and water in a redox reaction.

$2F_{2(g)} + 4NaOH_{(aq)} \longrightarrow O_{2(g)} + 4NaF_{(aq)} + 2H_2O_{(l)}$

Chlorine disproportionates with hot concentrated sodium hydroxide solution to form sodium chloride, sodium chlorate(V) and water

$3Cl_{2(g)} + 6NaOH_{(aq)} \longrightarrow 5NaCl_{(aq)} + NaClO_{3(aq)} + 3H_2O_{(l)}$

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Task c)

Chlorine is widely used in water purification, manufacture of PVC. Chlorine is toxic when inhaled in and strongly irritater/corrosive to the eyes, skin and respiratory system and high concentrations can be life threatening
Mitigated by use of closed systems and good ventilation, wear goggles, glover and suitable propective clothing when dealing with chlorine

A

M

Fluorine is used in manufacture of flaorinated compounds eg calcium fluoride. ✓ A
Fluorine is extremely toxic and highly corrosive, can cauce serve burns and seriour damage to the eyes and respiratory system. ✓ D.
Mitigated by used of closed systems and good ventilation, wear goggles, glover and suitable propective clothing when dealing with efluorine. ✓ M

A = Application
D = Danger
M = Mitigation.

AB Ae = application Explained.
Ae

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item 5

Task (a)

Benzene is a non-volatile polar solvent because there is a smaller difference in the electronegativity of carbon and hydrogen hence dissolve non-polar organic compounds ✓Pe

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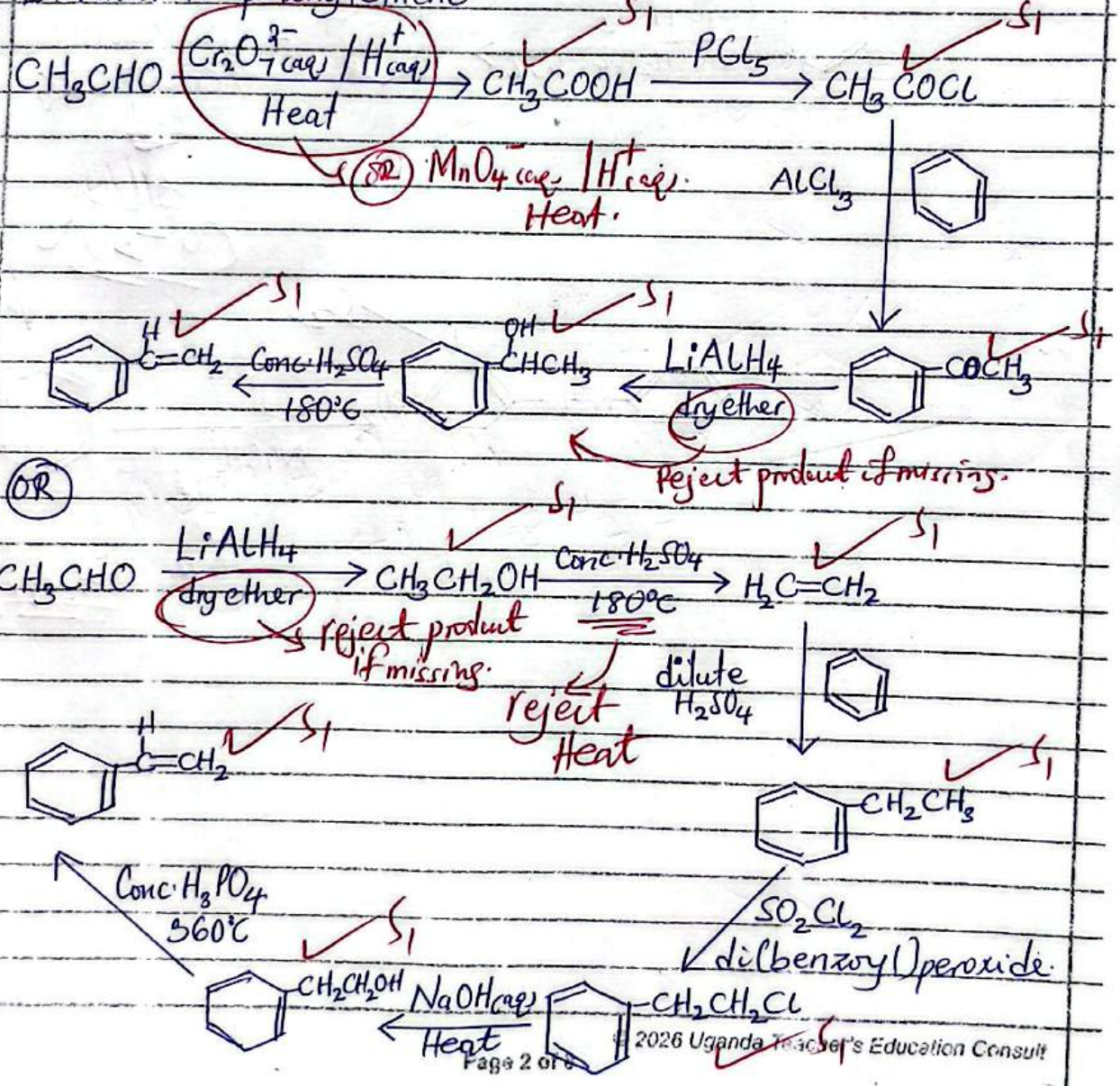
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item 5

Task (b):

Ethanal to phenylethene



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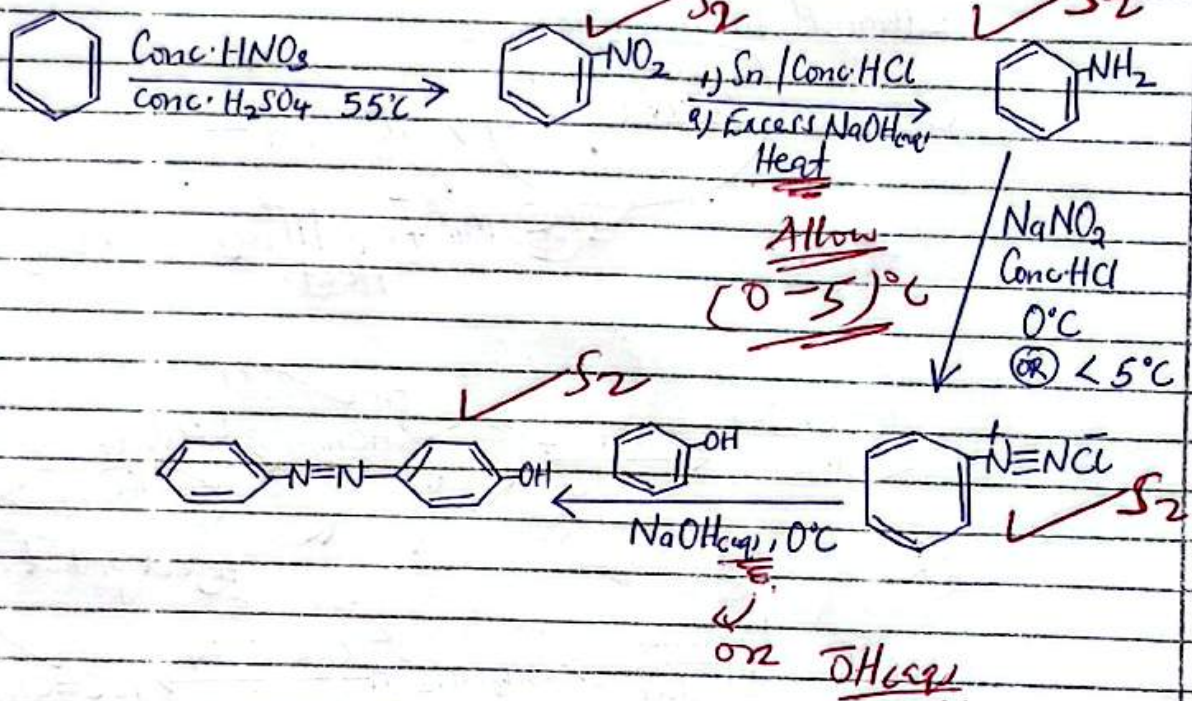
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items task (b)

Formation of the dye From Benzene



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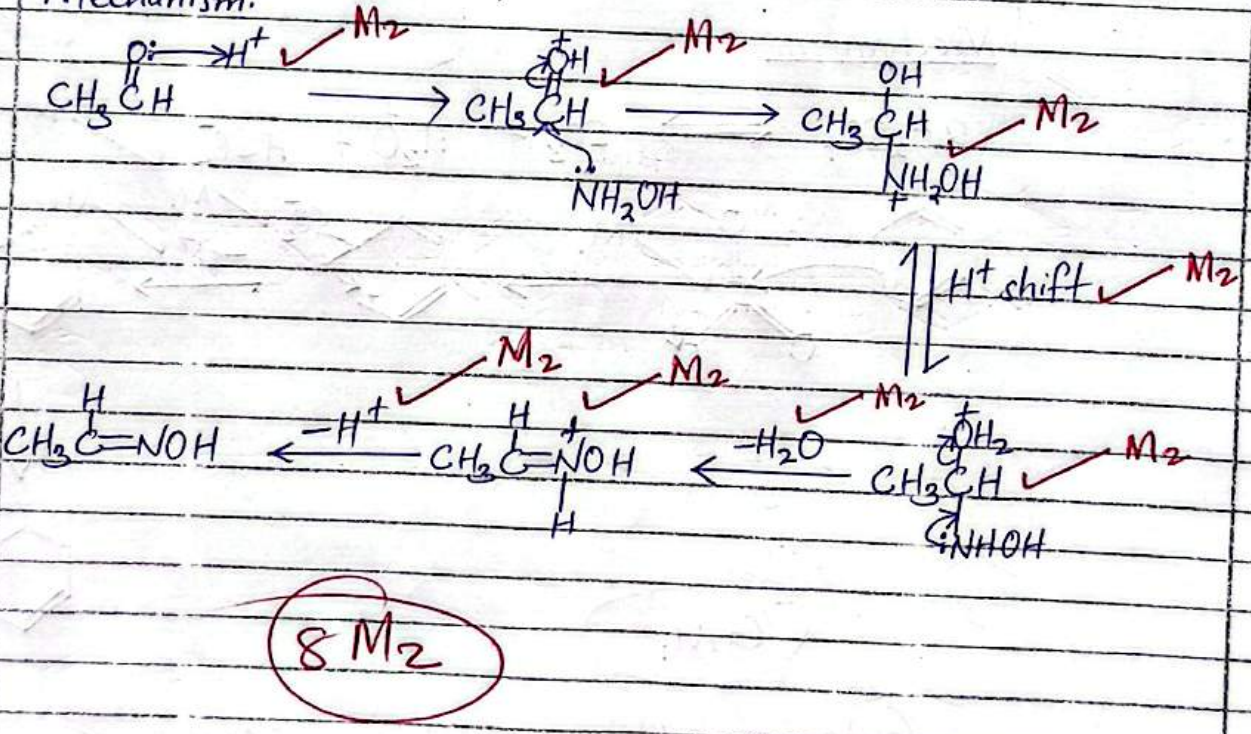
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Task (b).

Reaction of ethanal with hydroxylamine.

$$CH_3CH=O \xrightarrow{NH_2OH/H^+_{(aq)}} CH_3C(H)=NOH + H_2O$$

Mechanism.



8 M2

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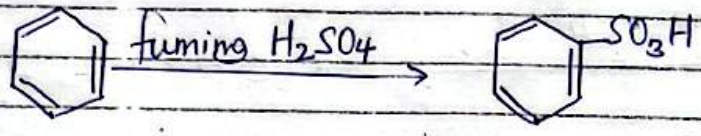
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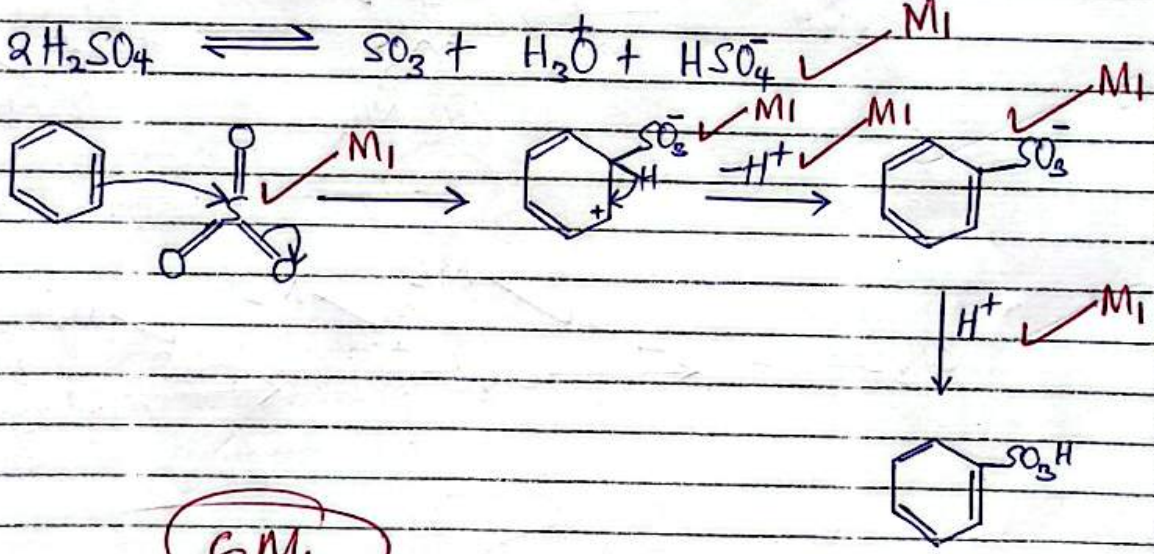
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Task(b)

Reaction between benzene and fuming sulphonic acid.



Mechanism.



6M1

6M1 + 8M2 = MAX SCORE

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item 5

Task (a)

Benzene is nonpolar and therefore used as a solvent for many manufacturing industries for mainly non-polar substances. ✓ Ea.

Ethanal is readily oxidised and soluble in water thus ethanol is important in making useful aqueous chemical processes. ✓ Ea.

Dangers of benzene.

Benzene fumes when inhaled cause blood cancers since benzene is highly carcinogenic, triggering uncontrolled cell devision. ✓ Ed

This can be migated by wearing face masks to prevent inhalation of fumes. ✓ Em.

Ea = Explained application.

Ed = Explained danger.

Em = Explained mitigation.

For

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Task (a).

Propene is a non-polar hydrocarbon with very weak van der Waals forces of attraction, exists as a gas at room temperature. ✓ Pe

Ethylethanoate is polar with dipole-dipole moments, without hydrogen bonding thus highly volatile. Pe

Ethanol and propan-2-ol have the -OH group with very strong hydrogen bonds which are hard to break thus less volatile therefore exist as liquids at room temperature Pe

Ethanoic acid associates through intermolecular hydrogen bonds, and forms dimers in liquid phase thus least volatility. Pe

(Pe) = Physical property Explained / Use Explained.

P - Physical property
E - Explanation. } (Pe)

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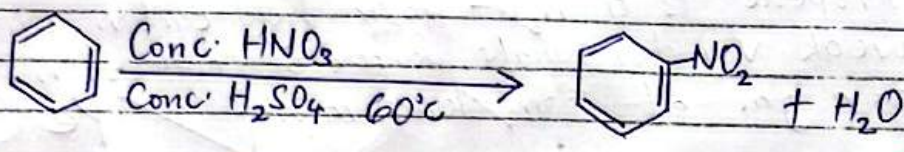
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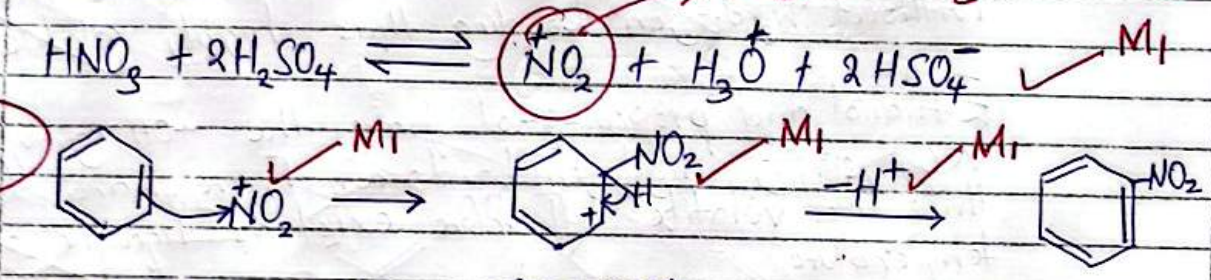
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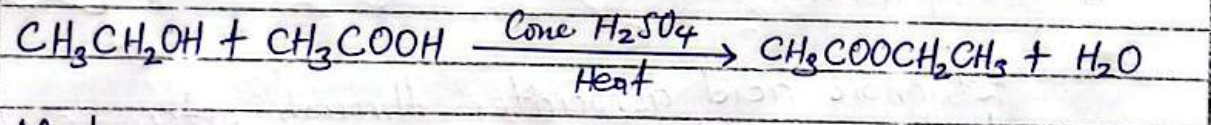
Task (b)



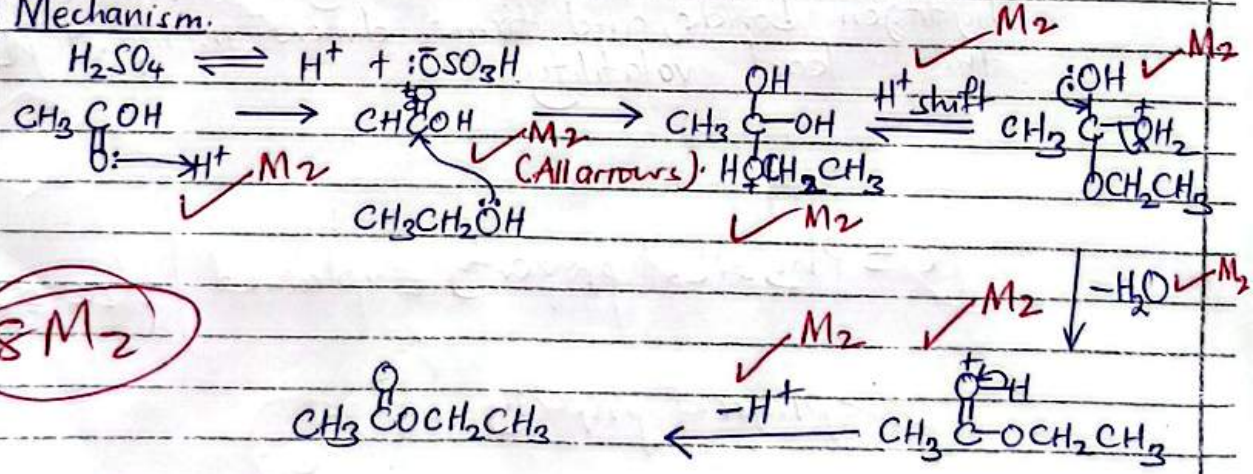
Mechanism.

Reject M_1 if NO_2^+





Mechanism.



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$4M_1 + 8M_2 =$ MAX SCORE

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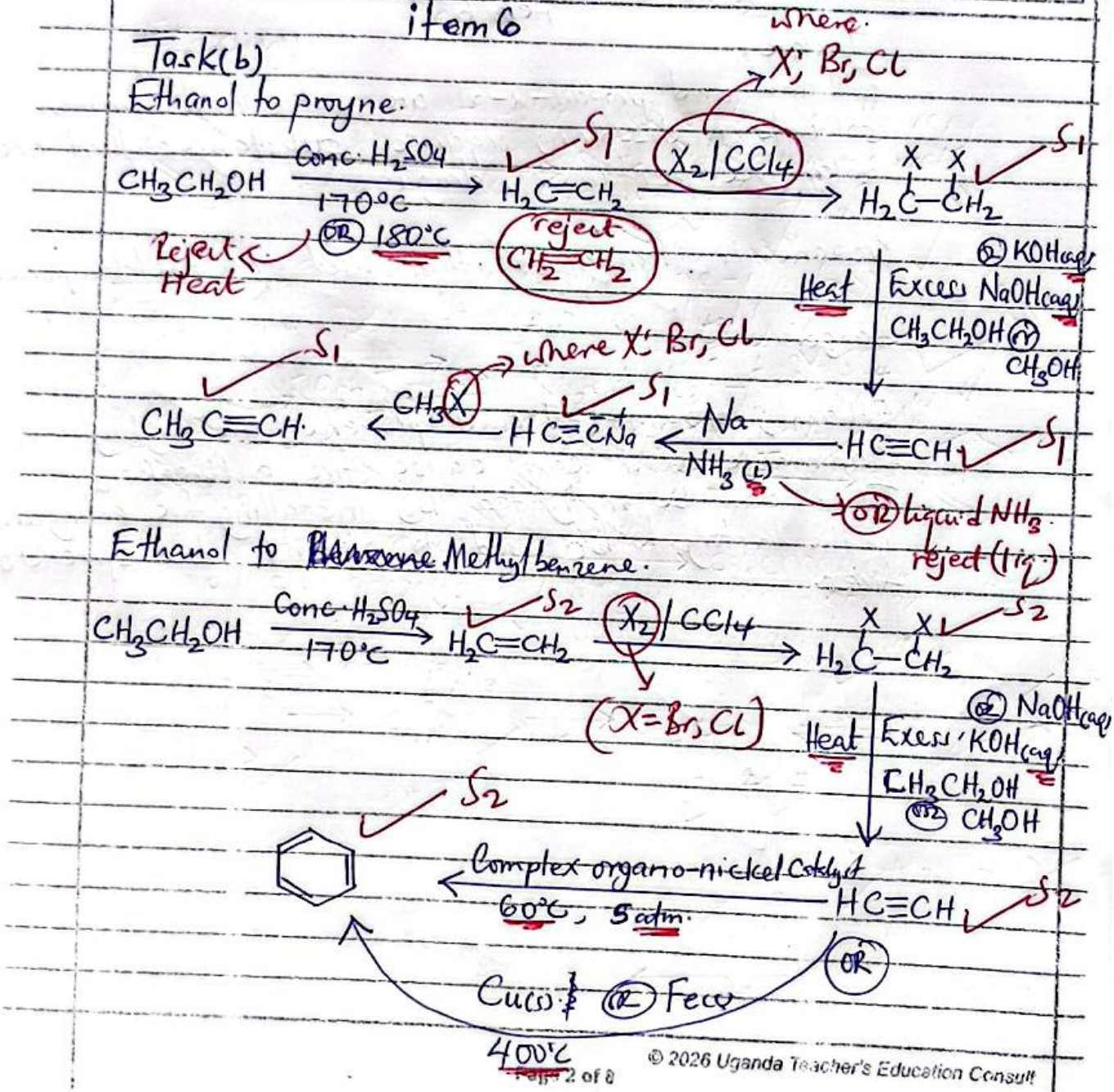
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Task(b)

Ethanol to propyne.



Ethanol to ~~Benzene~~ Methylbenzene.

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Task (c)

Ethanol and propan-2-ol are polar and common solvents in dissolving organic substances that are polar like perfumes, drugs.

Propene is flammable thus used as a fuel component.

Dangers:

Ethanol, propan-2-ol and propene are highly flammable and can cause fire outbreaks. M.
This can be mitigated by installing fire extinguishers in the factory to be used to prevent the spread of fires.

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